

ActInf Livestream #005.1: "Multiscale integration: beyond internalism and

Livestream | 1:43:45

hello and welcome

everyone this is the active inference

live stream

it is september 29th 2020 and this is

our episode 5.1 welcome

to team com whether you are a first-time

listener

participant or whether this is not your

first time

we are an experiment in online team

communication and learning related to

active inference

you can find us on twitter at

inferenceactive you can find us

on email activeinference gmail.com

through our public key based team or

through our youtube channel

this is a recorded and archived live

stream

so please provide us with feedback so

that we can improve our work

all backgrounds and perspectives are

welcome here we really want to hear

about

everyone's individual experience and

also as far as video etiquette goes

please mute if it's noisy in your

background

and remember to raise your hand it's

awesome that we have so many

participants today

and so to hear from everybody let's just

make sure to raise hand and all

keep um a cue of people who have raised

their hand

so today in act m stream 5.1

the roadmap for our discussion over the next hour and a half will be as follows

first we'll have a warm-up section where we give a quick intro and check-in

then we'll be discussing the paper we'll talk about the goal of the paper

we'll talk about the road map or the section headers

go through the abstract and then go through all the figures

we'll also be able to touch on some background concepts if we have time and address some excellent questions submitted by sasha

and other participants and next week on 5.2

we're going to have further discussion on this same paper

so please remember to save and submit any questions

that you might have that don't get addressed in this first discussion

awesome so here we go into the introduction

section please introduce yourself and your location and just say hello to everyone in a short introduction

and then pass to somebody who you haven't heard introduced themselves yet

so i'm daniel friedman i'm in daviess california

and i would like to pass it first to alejandra

you're muted yeah well

hello everybody um well i'm

in mexico uh specifically in guadalajara and this state where i'm living now um

but well i mean my professional
background

i'm i'm a full-time associate professor
in cognitive and behavioral sciences at
the school of psychology
at the unam and also i'm an active
member of the community robotics lab
at the wyom mexico also and well
my research focuses on perceptual
optimization processes
such as sensory acceleration and sensory
enhancement
and also the computational modeling of
perception

cognition and i'm like
uh hooked with the predictive processing
active inference
quality coding and energy principles
since long time ago uh it's like um
my passion um and i'm
like giving a full time some
some of my days to the breathing and
trying to understand
um this this particular free energy
principle

so yeah that's me
let's pass to alex

hello everybody yeah my name is alex i'm
from

moscow russia i'm a researcher in
system management school and trying to
find a way

how to join concepts from system
engineering

and active inference frameworks

so let's say for sasha

hi thank you my name is sasha i'm a
neuroscience graduate student at davis

and my background in
a free energy principle is just reading
and
enthusiasm for figuring out how to apply
this to
my own work with developmental
neuroscience
and just looking forward to the
conversation
i'm going to pass it to mel
hello yep can you hear me awesome
perfect hi i'm mel andrews i'm
in cincinnati ohio right now um
i have a background in cognitive science
and i've always been fascinated by
issues in theoretical biology
and switched to philosophy for my phd to
really focus
on um sort of what are called um
life-mind continuity theories and in
particular i'm focusing on
the free energy principle as a formal
articulation of
a like-minded movie
oh i'm gonna pass this on to
richard thanks uh so my name is
richard niemeyer i'm an assistant
professor of sociology in denver
colorado
in the united states um my interest is
mainly in understanding how we can
integrate
neuroscience cognitive psychology and
sociology into more integrated framework
um most of psychological or sorry
sociological and criminal chronological
theories drawing on a lot of behaviorism
from the 70s and 80s

it's kind of its theory of mind and how
the mind society
relates to one another so i'm trying to
figure out more up-to-date ways to
integrate mind-body and society
most of the mathematics that i'm
familiar with in terms of kind of how to
do that is a lot of fractal geometry
graph theory dynamical systems theory
which is really fascinating but i've
been really interested in kind of the
free energy principle
it seems to do a much more kind of
intuitive
fluid dynamic way of integrating
different levels of analysis
and integrating specifically cognitive
emotion action things like that which
are really critical when you get into
kind of that social level of reality so
i've tried my best to work my way
through a lot of papers by myself and
it's been a struggle so
to find a community who's kind of doing
this together and can kind of listen to
other people talk about these ideas
is is really really helpful so i really
appreciate the opportunity to be here
i'm gonna pass it on to muddy
oh classic it was on me hi guys my name
is muddy
so i was a i was a phd student with carl
briston
graduating back in 2016 but i also was a
medical student on the md phd pathway
so my interest was in applying
free energy principle to study
stroke dynamics non-invasively using

neuro imaging

i since left the the medical world i

practiced medicine about

a week and then uh went into the tech

world i worked for

google deep mind and started the medical

tech company which was

acquired by the company i currently work

for now versus where i'm i'm the

director of ai and we're really

interested in spatial web technologies

spatial web protocols and building

provenance

ais that are not only explainable but

action-based and adaptable so that's

kind of like

that and um i will pass it on to

professor cliffinger who is quietly set

there

thank you honey uh hi everybody my name

is john clevenger

um i am now well i'm in northern new

hampshire

jefferson new hampshire farm but

actually i'm generally out of cambridge

mass

and um with the mit media lab the city

science group and i've been at mit in

the human dynamics group for a number of

years

um my interest actually uh i'm

a full convert recent convert to pristin

la

the the first and cult as they were but

uh

continued reading the papers that are

coming out the productivity of this

and working uh with maxwell and casper

about how to apply this to
uh modeling um modeling new concept of
firm new
concept the market new concept of
community um
so we have some projects that we're
focusing on that and actually
collaborating with
muddy on that as well
cool let's go to shannon
hi i'm shannon crooks i'm in south
dakota currently but i'm usually based
at the university of california in
merced
um and i'm in a sensory motor
neuroscience lab
i investigate the neural dynamics of
music cognition
and specifically how we process rhythm
and then interested in how
predictive processing approaches can
help us understand
how we perceive rhythm and music i'm
also interested in
how groups of people interact when
they're experiencing music when they're
playing it
or listening to it and in larger
dynamics of crowds
i'm excited for today's discussion um
specifically
for how you know a crowd could have a
different markov blanket
based on its interactions throughout
maybe a musical performance or a
sporting event
cool i think we have maxwell
muted maxwell sorry about that

hey i'm maxwell ramstead i'm a
postdoctoral fellow
at mcgill university and i'm one of the
main architects of the multiscale active
inference formulation
that we'll be discussing today and the
first author on this paper
multi-scale integration beyond
externalism and
beyond internalism and externalism and
i will pass it off to alex kiefer
uh hi everyone i'm alex um as nestle
just said um
i'm coming from a philosophy and
cognitive science background um
i got my phd at the cuny graduate center
um
currently um at monash university in
melbourne but not actually there because
of the pandemic so
i'm in the new york city area um and i'm
interested in
uh i'm coming from sort of a traditional
philosophy of my perspective i
increasingly got into like connectionism
machine learning
modeling eventually the fep and i'm
interested in how this stuff ties into
fairly traditional representationalism
on the one hand and sort of physics on
the other and how these come together
so that's my angle and this and i'm
really excited to talk about this paper
cool i think stephen and then if daniela
jumps back in we'll go to her but
stephen i think you're the last one here
oh okay hi i'm stephen civit i'm in
toronto

and i do community development projects
using theater and immersive
participatory approaches and
ended up developing a process called
social drama topography
in south africa in 2006 and trying to
scale that process
in different ways has looked at
how embodied cognition fits into this
kind of
process of engaging communities to make
embodied maps together around issues and
that's kind of translated over time into
other sort of immersive practices and
i was looking at embodied cognition over
a few years and i kept
finally seeing this work with carl frist
and i thought oh my goodness i've
already
gone so far down all these avenues and
i've seen this other thing and then
suddenly i found a page which
spoke to how many citations he had and i
thought wait a sec and i suddenly
saw that his work was actually bringing
all these different threads
together into this unified framework so
yeah i'm really excited by that awesome
well
thanks so much everyone for coming out
to participate
um it's really a spread of countries and
levels of experience and fields of
expertise so it's about bringing it all
together and coming together around
these topics
so just as a warm up and to hear from
everyone who would like to speak by

raising their hand
let's just start with asking what drew
you
to this paper or topic today
so all star and while other people are
raising their hand and i described this
a little bit in the 5.0
video but i think the idea of
integrating multi-scale systems
is really um interesting and there's so
many ways to go about it
and it's almost undeniable that we're
living in multi-scale systems but the
question is
what now and so i'm always interested to
hear about
systems for thinking about what now and
also for different people's perspective
great muddy and anyone else can raise
their hands
muted though with a video off it's the
fine line between
muting the background noise and being
ready so i understand but go ahead muddy
my apologies yeah i think um nested
systems are
an incredibly important way to to view
real dynamical systems that we see all
the time in the spatial web so
we constantly have streams of data
coming from iot
and then we're kind of met with the
scalability challenge of how do we
actually model a system
accurately to represent the dynamics
that we that we wish without
getting bogged down in this sort of
field of complexity so by abstracting

away

these these layers into really really

meaningful nested

heterarchies is is really really key in

being able to practically

deploy some of these you know novel

learning paradigms in real world spaces

so i think without this sort of

quote-unquote dimensionality reduction

i mean that in more of a philosophical

sense than a mathematical sense

we're going to struggle to apply ai at

scale so that's why i'm

i'm particularly interested in the work

that maxwell and the free energy

principle group is doing

cool let's hear from maxwell then sasha

and then i also see other people but

just those two next

maxwell also muted

sasha do you want to go first i don't i

don't have to go next necessarily

sure i'll uh briefly say a bit

um i was drawn to this paper because

as a neuroscientist i think a lot about

multi-scale

um levels and

to decide at which level a system should

be studied because

even though um we're studying something

at a molecular level or even in a

different species

we ultimately want to make claims about

larger cognitive processes and how to do

that without abstracting away

really important aspects of the system

at the scale that you're studying it

it's a huge question

so um i really like thinking about these markov blankets at different levels and where to draw the line cool maxwell then richard then alex k i think my remarks are gonna echo a lot of what's been said um i got into the free energy principle stuff originally uh because i was looking for a theory that could address dynamics at and across multiple different uh well the way i used to see it is ontological levels uh in working with the free energy principle i've come to reject the traditional metaphysics of uh of ontological levels uh in favor of uh you know this view that there are big things and there are small things and uh big things are made up by small things and big things tend to be slow and small things tend to be fast and what you actually have is like this nested series of systems that are made up of increasingly big things um and yeah back when i was starting up my phd in 2015 uh there wasn't any theory that i could see that was really able to not only define dynamics at each of these different levels but propose a way a mechanism by which they were all integrated and this is what i found uh via the free energy principle so i i sort of made it my thing uh from 2016 to

now actually to explore the multi-scale
uh implications of this framework
awesome
richard then alex k yeah i think um
one thing that i've been really drawn to
the last couple of years is the
mechanistic
philosophical perspective of of carl
craver and lindsey darden
they give you this really nice i think
tool set on how to talk about
decomposing mechanisms levels of
mechanisms
which is very intuitive and i think
works well with with sociology
and biology which has this very clear
conceptual
way to move up and down quote unquote
levels of reality
but the thing that's i think always been
missing is how do i take that conceptual
language
and match it with an empirical framework
that allows me to
um model those levels of reality in a
way without losing any of the subtleties
and
any of the the conceptual apparatus that
is so critical to
talking about those those different
frameworks and the more and more i get
into the free energy principle it seems
like it can give you that empirical
language
that has a very nice natural transition
into that conceptual language i'm so
used to
that gives you all of the stuff you're

trying to talk about conceptually but
then
also gives you these empirical tools
that allows you to go above that and
talk about the organization of the
mechanism
in a very nice precise empirical
language um that's just impossible to do
i think with a lot of existing methods
in the fields that i'm used to
cool let's do alex k and steven and then
i'm gonna go to the second question
um so i i just recently sort of got
into this whole multi-scale picture um
associated with the fpp
and um i guess my interest in this is
um it seems to me like like i'm i'm
nearly 100 convinced that this is a
really good description of reality at
multiple scales but
i'm not completely convinced that this
has the radical kind of philosophical
implications that
that some people think it does i mean
this paper actually strikes kind of a
nice middle ground but
um i guess to put it briefly like i view
this as
we've got nested multiple systems each
of which
possibly has a physical as well as a
cognitive description and maybe the
bound
the bounds of cognition within that
hierarchy are maybe
less strict than may have been thought
like i'm happy describing cognition
or ascribing cognition to systems at

various levels within this hierarchy but
but i don't see that it's so hard to pin
uh
the cognitive properties of one
particular system um
onto a particular physical system so i'm
not i'm not sure
anyway i hope that that gets we can get
through some of that
also i'm not sure which system i am in
this hierarchy of embedded systems that
you know my brain involved many systems
in itself and that's
self-embedded in larger systems so i'm
trying to figure out
which type of system within that
hierarchy an individual person might
be identified with cool stephen and then
we're going to go to the next question
to muddy and mel or anyone else who
raises their hand
yeah i'm very interested in how this
gives a kind of action orientated
miso level approach which is really
useful
because when i'm working with say a
community trying to understand
how to connect to the individual to the
group
to the team to the organization or maybe
influencing change at the municipality
or
broader social levels there's this real
problem with bridging scale
um and it ends up getting caught up in
all this languaging that
whichever field has been you know the
psychological perspective or this

perspective on that perspective
so this gives a kind of seems like a
very neutral
clean um way of understanding
how people make sense of different
levels
and that's really valuable i think
that's been missing from
community psychology is really actually
psychology itself suffering from that
because it's kind of stuck on this
level of um perspective or
interpretation so this gives a way to
actually go into the sub-paragraph so
for me
getting into all of this has actually
opened up all this sub personal
neuroscience site
which i actually haven't really got into
before so it's kind of interesting
nice thanks for all these responses
also raised her hand yes so next on the
stack is muddy and then mel and feel
free to answer either
the first question or the second
question about what a multi-scale system
you work with is so muddy first
um great so they actually kind of uh
fall into each other so one of the
things that i wanted to mention was
um the practical challenges of
implementing
um inference in an industry right which
we kind of do it in a black box but i
think it was sasha who mentioned that
or alex who mentioned he had an interest
in this as well
um and what we find is that the the

practical correlates uh
in terms of hardware and chip design
that we need to to deploy this stuff
in reality is is difficult
based on the data sets that we collect
so one of the systems that uh
we're currently modeling right now is
warehouse pick and pack working for for
amazon
and understanding that
the the systems that drive why people
pick certain
boxes from certain shelves for example
is like a multi-nested
system of everything from how valuable
an item is to how tired the pick and
pack worker is even down to the function
of
how tall they are versus how high on the
shelf the particular item on the waiting
list is
so understanding how to pick that apart
and um
deliver value essentially in industry
eventually so yes we have this wonderful
way of viewing the world
but we also have a practical set of
tools which are a wonderful way of
modeling the world as well
and so having this nested system i think
is really really important
in terms of biting into the problem of
practical deployment of these
these types of models at scale in order
to deliver value now i don't just mean
business value i mean
you know inside value as well so those
are just a couple of comments that i

wanted to to introduce into the
discussion as well
so thanks for that mel and then anyone
else can raise their hand
hi um i was drawn to this topic because
um i've been trying to
convince maxwell for a long time that
the fep
is just um that basically we should
we should be taking as instrumental as
an approach to this
as as we possibly can this is like the
bare bones uh
it's it's just a model it doesn't
constitute knowledge of the world it
doesn't
constitute falsifiable claims it doesn't
constitute
uh et cetera et cetera et cetera it's
just a model and we just
we use it to get at things in the world
and and and that's it
um and this paper's uh interesting to me
because
in it maxwell is the one pushing the
instrumentalist line that that i don't
want to
bite uh namely that's a
that there's nothing more to
the boundaries of systems
than is it useful to model it that way
right is it
is it can it can it be usefully modeled
as the battery is here
for me i'm i'm very um
i'm very willing to privilege the
organism
as a privileged scale

or level or unit of analysis and i'm
very willing to privilege the brain
as well so in that order
interesting fighting words mel um
shannon and then anyone else can raise
their hand too
i i feel partial to those those last
words from mel
on privileging the organism or the brain
um but that could be just because i
i study the brain i want to know the the
neural mechanisms
that are giving us
any of our perceptions or engagement
with the world
um but i think i was drawn to this topic
because um i was drawn through
predictive processing literature and
like um andy clark's work and then
eventually on here to the free energy
principle
um but a lot of what was written
and like is now has these great
mathematical models
um that we've touched on before like an
ability to translate
um maybe some sort of phenomenological
experience
into a formal model um it meshed really
well
with my studies in music and what
like teachers of music will instill upon
you in that
you're basically
through your musical experience you're
modeling something about the human
experience
um so you're choreographing aspects of

tension

and release which sounds a little bit

like

balancing um predictions and

prediction errors um and you're trying

to maintain a certain boundary within a

musical piece

of the emotion that you are experiencing

or that you want your

audience to experience and this sounds a

lot like a boundary on a certain

phenotype

of experience um and just sort of this

like superficial matching of the

language of musical expression and then

this empirical

um like statistical language is

sort of what drew me in and i suppose

that answers the multi-scale system i

work with um

music and musicians and people cool

very interesting we'll go to richard and

then the third question

yeah kind of building a little bit on on

mel's point

in terms of the multi-scale system so

obviously as a sociologist i deal with

groups i deal with society

and you know i can take an individual

and i can take two individuals now i

have a dyad

i have another one i have a triad i can

model that as a network

i can collapse the network down and

model it as a dynamical system

and now i can talk about organizations i

have multiple dynamical systems talk

about them interacting

now i have a network of coupled
oscillators i can go in i can use
fractal geometry to talk about how
properties are scaling up and down
from the dyad the triad all the way up
to that
network of coupled oscillators and at
the end of the day i i'm never really
confused about whether or not i'm
really talking about networks or am i
really talking about dynamical systems
they're all just models that allow me to
capture the phenomenon that i do believe
is real i do believe society exists
if you ever want to get into a very
strange conversation go to a conference
of sociologists arguing about whether or
not the united states is real
and my comeback is always don't pay your
taxes on april 15th
and you'll figure out very quickly if
the united states is is real or not
and so i think you know a lot of these
philosophical questions are always
fascinating
especially at three o'clock in the
morning in a pub when you've had too
many to drink
but i think ultimately in the day you
know there is a concrete reality
the question is what empirical model
just allows you to model it
as true to the theory or the concepts
that you're that you're dealing with
um without losing information you know
unnecessarily so i think
you know again i think any empirical
model that allows me to do that

is fascinating um but at the end of the day i do believe you know the humans that i'm ultimately talking about are concrete and are are real nice i'm gonna introduce the third question we'll go to buddy first here and anyone else can raise their hand the third question is why does our philosophy of systems matter um that is quite bizarrely scary and psychic because that's what the point i was going to make just to just to piggyback on on richard's comments just there which is um i remember these discussions these really really lively discussions about you know how representative is this model of reality and i remember at some point just thinking well how useful is this a question to ask you know at some point is is the model just a tool a means to an end and do we employ bounds on what we expect from these models in terms of their utility and if these models and their outputs satisfy these bounds great you know i i sometimes i i see some of the the discourse around free energy principles and and what it means philosophically and we lose what it gives us in terms of explainability and an actual measurable dynamics so i actually

fall probably like very far on the side
of

i don't think our philosophy of systems
matter at all i think our explainability
of

empirically measured data is by far the
most important thing

and trying to explain systems where our
data sets are incredibly sparse

so for example the entire hierarchical
organization of the brain from cell to
effective connectivity

i think is um sometimes missing the
point of just how valuable

these novel uh schemes can be so that
that's kind of what i wanted to touch on
that

nice stephen and then anyone else before
we close out this question

yeah i think that that empirical pieces
and practicality or
practice application is really a good
point

one thing that has caught me out though
and i've realized is

is that perspectivism is very much
part of systems generally and it's also
part of psychology like we take a
perspective on stuff

and there's this assumption that
everything well what's your perspective
that word just gets used all the time
and it's like but the thing is that the
free energy principle gives you a way
and then they say okay and now we're
going to do stuff that's performative or
arts based

and it's a bit like we don't know what

to do with that so we just call it
another field
right and then they do that but the the
the the active inference gives you
realize well there's
there's a way that you can abductively
or from
within be creating meaning
um during experience and it's um
i suppose a bit like that person picking
stuff from the shelves and they're not
sitting there as they do it having a
perspective like they may
take a perspective if you give them a
qualitative survey later which is what
nearly all social science does so you
take us perspective a first-person
perspective on your experience but the
experience wasn't first-person
perspective the
experience was just the experience
so this gives a way for that to all be
kind of um
unified in some way cool let's go to
alex and then richard then close out
this check-in
um just really briefly i mean i think i
think it matters philosophy systems
matters because
i think the philosophy well i guess i'm
sort of a holist
in a way at least in terms of
explanation so i think it matters
because
um the philosophy of every little bit
matters
uh because it connects to everything
else but also this is sort of a question

that
can be applied at multiple scales so so
it could potentially matter a lot
i guess i mean that doesn't really
answer the question of why philosophy
matters ever
but i think it's because it's part of a
big theoretical framework
i'm sort of acquiring about this stuff
so these questions about realism
and instrumentalism don't land all that
well with me because
i mean for example i think the fep if it
ends up being part of our best overall
sort of account of things
then i think it's confirmed or
disconfirmed
by proxy whenever anything any
observation is
is recorded right so um
unless you want to say that it doesn't
have any inferential connections to
anything else in which case
it's really weird i don't know why we're
talking about it um
anyway that's enough for me perfect
richard and mel to close out the section
thanks
yeah i think in many ways and this is
probably just maybe disciplinary
specific but the philosophy of systems
and the philosophy of
what we do in science i think is
incredibly important in the social
sciences because
so much so much of our concept so much
of our
our theory that that we deal with now is

is deeply indebted to
dewey and uh coned and you know hegel
and these these
uh philosophical arguments about the
relationship between mind body and
society
what is a thing what are the boundaries
of things and
these things have been baked into our
theories and many times we don't
recognize that they're there and they're
they're deeply affecting concepts and
how things are operationalized and what
we consider to be a truth value or not
and to go in and unpack those theories
find those philosophical arguments
and then interrogate them from these
kind of emerging perspectives and say
you know was whitehead onto something
about kind of you know
uh the extension of reality as an
ontological category or was he not
and if he wasn't and this new principle
can deal with a lot of those ideas
in ways that were significantly shaped
uh sociology
then great we have something to kind of
move on to um so i think that
philosophy is always this thing as much
as i try and get rid of it it's always
coming in the back door some
somehow some way and i think just being
sensitive to it being comfortable
dealing with it and interrogating it
from these new perspectives
i think is always going to be something
that's very important because it's
always there in the background

somewhere if you dig deep enough thanks
well said and then
mel and then we'll let maxwell have a
last stop as well
in my mind philosophy doesn't answer
questions it raises questions
um and the empirical sciences attempt
to answer those questions
but you need philosophy but don't chew
on the computer jesus christ
you sorry you need philosophy to both
raise those questions
and then to uh to tell you
how those questions have been
potentially answered
by the empirical results so
um there is no such
thing as as kind of science operating in
a in a vacuum without philosophy of
science that's my opinion
uh that that there's this necessary
dialectic between the philosophy of
science and science itself
um going on
cool agreed maxwell and then professor
clippenger if you'd like
um well so i think that there's a
there's a methodological component to
this as well
i i want to echo what what everyone said
earlier
uh you know if if we're gonna study uh
multi-scale systems it's good to get our
ontology straight
and you know the fact that nature seems
to have this multi-scale
structure suggests that our theoretical
accounts of nature should also reflect

its uh multi-scale structure
but i think from a more pragmatic point
of view
um i i like to think that a lot of the
uh kind of single discipline uh topics
have
uh been covered in in a very significant
way
but that there isn't there isn't the
same kind of
um you know established work
connecting different methodologies that
operate uh
at different scales so i think that you
know the
getting this systems take right not only
has
theoretical implications but also has
practical implications
uh for the way that we conduct uh
research namely
that we should aim to form uh
multi-disciplinary groups
that address questions of interest from
several different
angles simultaneously so and this is
echoed in the paper
that we'll be discussing today where
what we're essentially trying to do is
advance
uh on the one hand a pluralistic
ontology a multi-scale uh ontology and
on the other hand
uh a a uh methodological pluralism
uh where several different uh sets of
methodologies can come and interact
together
nice professor cliffinger or anything

else there

you're muted but yes um okay i have to
apologize because i have to leave early
but this is a great discussion a great
beginning i'd have to

i just want to throw my hat in this i
i'm sort of in in

a muddy's camp uh and looking at
explanatory power

um and i think a lot of that is there i
i'm so sort of

uh ditto with with what maxwell is
saying in terms of providing an
integrated framework

uh i i of course as a philosophical the
philosophy has a place in it but i think
we're there there were so many
losses embedded in historical situations
that are really not

that i think we're at another level of
understanding that

it's what becomes really important is
being able to start to test these things
and be able to rigorously test them and
because there are a lot of very
powerful claims that are being made that
if they're true they're quite

transformative but they

they need to be tested awesome thank you
everyone for this very exciting check-in
and i'll just add one

comment on why the philosophy of systems
matter and it's examples that are
brought up

by um helen longino in her book studying
human behavior

uh which level of analysis which
discipline we default to

molecular neuroscience or environmental studies if we default to a higher level of organization or lower level to look for our explanations philosophically it's going to directly translate to what kinds of interventions we think are going to be possible in the system and so if the story about for example aggression and sexuality which is what professor longinus studies in that book if the story is it comes from jeans well then you're going to be less likely to undertake affordances related to the environment if the story is um about the environment you're going to be unlikely to take affordances related to molecular interventions so even before we say what it actually is and that's of course a complex question or what is the best thing for the system i i would fall on the side that it really does matter in the background and sometimes it's just the water we swim in how we think about these multi-level systems so let's just like for a three second aside i i will happily admit that i'm a staunch empiricist somebody who worked in industry basically the moment i could and so i you know what i've i'm just conceding a little bit that actually

you're right without the philosophy of
science we don't introduce
novel ideas that are worth testing in
the first place right and
that discourse is what leads to the
models that allow me to try to issue
value from that
quote-unquote value from that so i am
backing down just a little bit from my
empiricism after this discussion
so i just wanted to say that yeah nice
um
a live conversion story you heard it
here first
but even pragmatism is defined within a
use
view so it's like all models are wrong
some are useful and then the
second part of that sentence might be
useful for whom or under
what conception there's no model that's
just useful outside of a
value set so pragmatics isn't just like
let's just do the most useful thing
there's not just one useful thing
so after that extremely interesting
check-in let's get to what the paper is
about
and what the goals of the paper are so
the paper
is multi-scale integration beyond
internalism and externalism
by ramstead kirchoff constant and
fristen from 2019
and what the paper writes their goal as
is here we focus on making sorry someone
needs to mute um
i'm going to mute that person here we

focus on making explicit a description
of the boundaries of cognitive systems
that we think follow from taking
seriously the inactive
embodied and extended nature of
cognition this is the idea
that the boundaries of cognitive systems
are nested and multiple
and that with respect to its study
cognition has no fixed or
essential boundaries and so it's all
about this taking it seriously with a
multi-scale integration
so how are we going to take the
multi-scale integration
seriously i'm going to run quickly
through the section headers and anyone
can just
raise their hand and i'll pause the
section headers
just to address the question so first
there's an introduction of the paper
which makes sense because there's so
many areas that are being tied together
here
there's the first figure which is about
markov blankets and active inference
in the second section of the paper which
is called a variational principle for
living systems
first they talk about this variational
free energy formulation
for living systems and show a figure of
how the free energy principle and
self-evidencing are related
to living systems they then talk about
generative models and action policies
because when we're talking about

internal
models or internal states of an organism
we're talking about generative models
and action policies
they then talk a little bit more
specifically in the text
about markov blankets and the boundaries
of cognitive systems
which recalling their goals they're
going to take us to somewhere where
these boundaries are
real and they're not equally prescient
as they say however
that doesn't imply that there's one
single correct level of analysis
this is related as maxwell also brought
up earlier
on the formal ontology for the
boundaries of cognitive systems
because the modeling approach the
descriptive or action-oriented modeling
approach that we take
to a system is related to how we think
about it and so that's the
background if your ontology has a
certain type of scheme
pre-built pre-installed then your models
are downstream of that
so when we're talking about philosophy
of science philosophy of multi-scale
systems
complex systems we're talking about what
ontology will let us
draw the boundaries around these systems
in a way that's yes
useful but also aware about these other
aspects
and the fact that we're not the only

perspective on use
out there this takes us to section three
on cognitive boundaries specifically
this debate
long-standing between internalism and
externalism
it ends with an ism so it's a dogmatic
belief
and it um is just what it seems to be
it's about whether the external
factors in cognition or whether the
internal factors in cognition
play a more relevant role or a primary
role
and the internal and the external are
implicitly at the level of the organism
and so
one of the i think strongest stances of
this multi-scale analysis
is well isn't there an internal and
external to the group
isn't there an internal and an external
to the enacted
ecological cognition like a human in a
building with a computer
what does internalism and externalism
mean when we're talking about
niche construction when we're talking
about sociological systems
and so to explore that first they talk
about
externalism in the context of radical
views of cognition
so radical means from the root and so
to take it from the root that
externalism is the nature of cognition
would be to kind of have a causal model
where the outside things are more

causative
going in rather than the other way
around in figure 3
there's a very nice abstract
representation of
operational closure which is actually
related to
kant and the idea of teleology and
whether the organism
is the fundamental unit of analysis in
biology or whether other levels might be
important too
and figure four shows from a autopoietic
um point of view about autocatalytic
closure
and so it's relating using a few
different metaphors including uh
autocatalysis like a metabolic loop
metaphor for closure as well as an
informational closure that will be
returned to
when talking about the markov blanket
internalism is then given their say and
it is pushing back against externalism
and
if one can imagine what radical
externalism looks like you can probably
imagine
what radical internalism looks like
and then after bringing up in section
two the variational free energy
principle and then
in section three framing this debate
about internalism
and externalism in section four we
combine sections two and three
which is to say that we apply the free
energy principle formalism

to taking it seriously integrating
multi-level systems
between internalism and externalism or
moving beyond it
and that implies re-analyzing
generative models what they are and how
they're used to study cognition
a section called inactivism 2.0 which is
a callback to
earlier uh generations of an activism
then there's a section on nestedness or
how to study cognition beyond the brain
so okay neuroscience or behavioral neuro
is about what's happening in the brain
but that's nested within a body that's
nested within a social group
so how does this nestedness change how
we study cognition in the brain and
beyond
and that's represented in figure 5 with
this blanket of blanket ideas
and in figure 6 with a multi-scale
self-organization and active inference
figure and then the paper closes with
a movement towards multi-disciplinary
research heuristics for cognitive
science
so just as far as today's discussion
goes in the next 30 minutes
we're gonna try to run through the
abstract and the figures and then we'll
have some closing
thoughts and then that will hopefully
get everyone excited and interesting
thinking about different types of
questions which we'll be able to address
in live session 5.2
in the service of the half hour we have

to get through the six figures
i'm just gonna put the abstract part one
up here
what they're doing is presenting a
multi-scale integration
interpretation of the boundaries of
cognitive systems and they're wanting
this to be a corrective
for the debate between internalist and
externalist positions
the second part of the abstract they
discuss how they first
survey key principles of this radical
extended or externalized cognitive
perspective
also implicitly in the context of the
internalist perspective
and then they develop a multi-scale
account building on the free energy
principle
trying to move beyond this internalism
externalism debate
and then in the third part of the
abstract they close up that discussion
by saying that their approach does not
privilege any specific given
boundary of the world but it also
doesn't argue that all boundaries are
equally prescient so that's going to be
something interesting to unpack
and then they close just by saying that
they
are drawing the boundaries of cognitive
systems
using this multidisciplinary research
approach
so let's get to the figures and this
first discussion let's just try to

understand what are the figures about
what is being shown in the figure just a
first pass
level visually what do you see but also
where does it fit a little bit
more broadly into the a to z of the
paper
or just what did you think about the
figure what did you see in the figure
or what were you wondering about so here
we are
in figure one would anyone like to
take a first pass at describing figure
one
what they see or um what it shows or
means for the paper
all right so i'll start describing it
and someone can okay max well go ahead
well so this is uh this is a depiction
of a mark called blanket
uh and so markov blankets are one of the
uh kind of building blocks or
ingredients
uh that go into uh the free energy
formalism um and
essentially a mark called blanket uh
well mathematically what it does
is uh if you have a set of random
variables
um and you want to effectively
uh map uh or individuate is
a random variable or a set of random
variables from another
set of random variables so for instance
um
you know you want to individuate a
system like the brain
from all the other uh you know dynamical

uh systems with which it interacts well
what you do is that you stipulatively
define
uh a third set of states uh between uh
the the internal states of the system
that you're
interested in discussing and the
external states
of the embedding environment this third
set of states is called a markov blanket
and it's defined by the absence of
connections
so effectively uh the
internal states are influenced by
sensory states but do not
influence them and um external states
are
influenced by active states but do not
influence them so by precluding
these two uh basically directed edges of
this kind of connected graph
what we're doing is uh essentially
formalizing the kind of
channel through which any interactions
between
uh the environment uh and the uh
systemic well the the internal uh states
of the system
have to have it has to go the interactions
have to go through this channel
effectively
um so uh yeah that's what that's what
the
formalism does for us it it allows us
to individuate
a system formally in terms of the
interactions that it has with this
with its embedding environment

great and
mel go for it
yeah i think a lot of a lot of people
publishing on on the fep and the markov
blanket formalism
but even more so in kind of the
discussions you get after a conference
people really expect of the markov
blanket formalism that it's
that it's picking out not um
token distinctions but type distinctions
so
so people really want the markov blanket
formalism to
cut at natural joints that is to to find
kind of the type boundaries between
natural kinds sorts of natural systems
um and it really doesn't do this
it really really doesn't do this it
really picks out individual systems
that's what it does
is it finds token systems
um and
naturally i think a a formal methodology
that that isn't about um
it's not about finding privileged scales
privileged
uh spatial scales or privileged time
scales it's
uh it's about finding particular systems
and so i think naturally when we employ
this formalism it's not going to lend us
a view
of um of privileged scales because
that's not what formalism does
cool and i think that's kind of against
goes against what people tend to think
the markov blanket formalism is for

awesome muddy um
i was just gonna say that uh so i would
say that most people
on this call are far more intelligent
than i am
and i i try to come up with with my own
sort of
baby interpretations of this which to my
knowledge nobody has
been angered by what and i think really
what this
tells me is if i have an agent
that is interacting with an environment
the only the only channel by which they
can actually
update their beliefs in order to change
their actions say comes through this
sort of border state these sensory
states so it's
it's interesting in that um i don't
consider this
well i mean it is a separation obviously
between two between two
um objectively different realities
should we say but it also
um it speaks to the the
force with which uh um
the the analytical link between the two
through
a certain set of sub states which is
which is really important
right and this is this border set on on
all your sensory states
so your sensory inactive state so that's
kind of how i've always
viewed this i have two sets of realities
that are trying to come together in
terms of

what's going on one of them has a belief
state one of them just is
and the the belief state is really
trying to figure out what is
and the only way it can do that is by
modifying its sensory state so that's
just

something which for maybe people who
aren't actively part of the community
might make it a little bit more
reductionist and simple to
to digest but if anybody disagrees with
me please

let me know nice well said um richard
and then if anyone else wants to raise
their hand

yeah so i'm sure this is probably a
naive question but you know when i was
listening to maxwell talk about it and
looking at this figure
do you do markov blankets work in the
sense of

i i hypothesize a priori that there's a
thing called an environment there's a
thing called an organism
there's a thing called sensory states
and then the markov blanket allows me to
go in and find it statistically and
determine whether or not these things
can be isolated you know empirically and
therefore kind of reinforcing my idea
that

these things actually exist in reality
or can i just feed a markov blanket
a whole bunch of random data and it's
gonna find something you know
kind of that garbage in garbage out
model so like

when you're when we're talking about
barcode blankets and finding these
boundaries
um how exactly does that does that work
let's have maxwell just directly respond
to that very nice question
and then we'll continue on the stack
with stephen and then mel
uh sure um to answer that question it
would be easier
if you could change figures and go to
figure
let me just check on the in the paper
it's figure five
sure so this is my favorite figure uh in
any of our papers
uh and yeah so this
this basically tells you uh what
what's basically going on here uh in the
construction of a markov blanket so
um in more recent work this has been
formalized in terms of two operators a
grouping operator
and um a uh
a dimensionality reduction operator and
effectively what we're doing is well so
uh to answer the question richard's
question directly
uh in principle uh what we've been able
to do recently
is uh develop techniques that
more or less automatically although you
know we can discuss there are a few
design decisions that have to be made
and
these have been discussed in a few of
our
discussion channels recently i'm sure

mel will uh
i'm sure you'll remind me of this uh but
basically what happens is
um what you do is you start with a bunch
of states right so we're working from uh
dynamical systems perspective
and we define whatever states are of
interest for us
and essentially the markov blankets are
constructed
by looking at the relative rates of
change of variables
with respect to each other so
mathematically what we're doing is
taking the
the partial derivatives of each variable
with respect to each other and
and then using effectively these
rates of change uh to construct the mark
all blankets automatically so
you start with a bunch of states uh here
i'm at the bottom panel in the figure
where you have essentially these nine
circles that represent states
so i mean these can be anything
whatsoever uh in a recent paper that was
pre-printed
this has been used to analyze fmri data
so in
in that paper each of your states is one
3d voxel
um and basically what you do is you
write down an adjacency matrix
uh so an adjacency matrix is uh
effectively a a a matrix where you have
all
of your states by all of your states and
the entries in the matrix are

quantifying the effect
the influence that one uh state has on
another
and if you take the if you take the
jacobian of this adjacency matrix you're
replacing all of these entries with the
partial derivatives of one
variable with respect to others and
uh so having said all that you you can
construct the markov blankets
analytically by noting the the absence
of dependencies in this adjacency matrix
so zeros in your jacobian mean that
the partial derivative of one variable
with respect to that other variable is
zero
uh which is another way of saying that
basically the two variables in question
are conditionally independent in that
direction
um and so i mean there are a few
modeling decisions
uh that you need to make effectively uh
there are most there are two to my
knowledge
one is basically uh going to be deciding
which states
are internal states uh
effectively so you're going to say okay
if this one is an internal state then
its blanket is this and the external
state's relative to it or that
uh that arbitrariness isn't all that
arbitrary in the sense that
the you know uh the blanket structures
are gonna be essentially the same it's
just what
what is what's going to count as an

internal state might change
um and then there are a few design
decisions uh when you're
when you're implementing this uh
basically you have to when you're
constructing your adjacency matrix you
have to kind of decide what counts as a
cutoff
for interaction so those two things
might make it less
objective but in principle if you'll
allow for these two kind of
modeling decisions you can feed these
algorithms
like a time series data for example and
it'll chunk out the markov blankets from
it
very nice answer maxwell thank you so
we'll go to stephen and then mel
then alex k and shannon
okay just get my head into it but um
one of the things that i think this also
is
maybe not always easy to imagine is that
is the dynamical
nature of all of this that it's talking
about states so it's not you're never
actually
being one thing you're always in this
state
of um unfolding
and that the markov blanket then
is that kind of statistical kind of
um you could say record keeping
at the different scales ultimately i
would say everything's at the
cell level to some level because that's
the kind of engine

that's enabling everything else to be
created the only thing i kind of
feel that is important to maybe
distinguish when talking because i think
there's a danger with the word systems
is that there's like i see these
blankets as more like a swarming
of um statistics which
creates the structures and the networks
such as the kind of
leafs or whatever and they and it's that
kind of structures which actually are
kind of in the world so to speak and
those structures have systemic
properties i.e
they become the um support mechanism for
a plan
the leaf does or but the structure has
come out from the swarming
of the um actual blankets
so that's just i think i'm still trying
to get my head around how to
pass it nice we can return to that
mel and then alex k
yeah i think i think maxwell basically
said it all but but
the the short of it is um
you have some time series basically and
that could be that could be um
real data or that could be uh sim
data that could be simulation right and
you're pointing the mark off blanket
formalism
you're directing it to the internal
state
and you're handing it your threshold for
for how conditionally dependent should
be conditionally independent enough

to count um and then
that does the rest of the work for you
that that that will partition
your state so so you give it the
internal states and you give it
the kind of threshold for conditional
independence and that partitions the
rest into external states
uh sensory states system active states
cool
alex k then shannon and muddy
yeah just a really brief point i'm just
wondering if one way of thinking of this
is that
the markov might get partitioned insofar
as its objective
um together with some data like maybe it
maybe it cuts nature
at particular joints but it just doesn't
cut at types of joints
in a way um so i don't know maybe we can
pick that up later
well yeah that's a good point if i can
just jump in here i think mel
is right when they say that uh
like the so it's it's the the markov
blankets are always going to pick out
individuals uh if there's anything like
a species level regularity that we can
express
or like a you know irregularities that
are specific to members of
members of the same group uh that'll be
more in terms of the generative model
non-equilibrium steady state density
uh so i mean you know this is also
discussed in the paper but essentially
what we're saying is you can imagine a

joint probability distribution all over
all of the variables of the system
and implicitly also over some fictive
external states that the system is
interacting with
and this entire joint density uh you can
think of as the phenotype of the system
uh and it's known as a non-equilibrium
steady state density
or a generative model so i think yes
things that pick out types have more to
do with this
uh joint density that tells you like
about the kind of creature that i
am whereas uh as mel was just pointing
out like
these are uh these these markov
boundaries are effectively the
boundaries of individuals
every time so cool shannon
and then muddy
i think i might be echoing a lot of what
was said already
um but on on what
the markov boundary is is is
separating um it's always going to
be a decision on the part of the
researcher for
for what correlations seem to matter for
the explanation
that you're looking for for the
particular system's behavior
so like if there's certain long-range
correlations that are
constraining short-range correlations
whatever your whatever you're wanting to
answer so if i want to answer
how um a

certain musical phrase affects the
movement of the person listening to it
the
particular the particular
um like enter the particular part of the
system that's interesting to me is maybe
not
going to be the electro electrical
firing of a neuron
at that moment because the electrical
firing of a neuron
at that really short range correlation
that really short-range time scale
isn't going to have a huge effect at the
long range time scale in my explanation
even though it's important you can't
have the reaction without
the brain having some electrical
activity
it's still not the most informative
scale of analysis
nice muddy and then mel awesome
i um so again just just bringing this
back to
uh a level of understanding that allows
me to like get some sort of intuition
why this hierarchical
uh markup framework is is really really
important and useful
in defining dynamical systems so one
example
you know my background in in medicine
reason why we
we were interested in functional
neuroimaging is we wanted to understand
at what scale can we
intervene in certain effective
connectivity architectures in order to

make clinical difference
so understanding the boundary states
of which pieces of this puzzle are
genuinely boundary stage you know
genuinely fall within some markov
blanket is really really important
and one example of this is if i take my
auditory course effects and my visual
cortex
at a hierarchical level of processing in
here
one could say that the internal uh
circuitry
that allows me to interpret you know the
the uh signals from my cochlear versus
my retina
are not correlated with each other
however at
an integrative state above this when i'm
trying to actually make some sort of
cohesive view of the world they are
both of course within that same markov
boundary state
so here's an example of a hierarchical
separation of boundary stage which is
really really important not only from
an empirical modeling decision in terms
of like what level do i abstract but
also
from an understanding perspective you
know one can
say assume a boundary state and then
test empirically if this
if this is correct would say some sort
of a dynamic causal model
and simply just by changing what you
think the actual
markup boundary states are or you know

the scopic level of that markov boundary
um you can start inferring things that
are really like clinically useful for
example
so the you know the motor cortex and um
the cerebellum quite clearly form a
markov boundary
for motor processing but then motor
cortex and s1 also form a markup
boundary in terms of action decision
making so
this tool it's it's really a beautiful
explanation of the world
but it's really a beautiful modeling
tool as well we're abstracting a way
hierarchy called noise in terms of your
model so i just wanted to add that in
oh i think that's a great point i mean
it it shows the kind of
meta modeling uh strategy that we have
going on here because
so uh in another paper that i think
you've discussed on this podcast a tale
of two densities
um uh i would i'm not sure that you did
uh
we've discussed a few anyway uh in
uh in
in that paper uh well actually
maybe we should just i'll i'll just seed
my my turn
sorry no all good all good let's return
to mel
and then anyone else can raise her hand
yeah just to quickly clarify um
contrary to what alex said i think um
the term
the the phrase cutting at natural joints

as it's used in philosophy of science
actually refers specifically to
to type distinction not not token
distinction so it's actually
it's about finding natural kinds and
natural kinds are
are definitionally type distinguished
and altered distinctions
yep also if i could add one note there
from my field of gene expression
analysis
there's often a few different ways to
construct a network of relevant genes
for
you could uh pull down and see which
proteins are physically touching or
which transcription factors are binding
to which genes so that might be a
protein-protein interaction network
or a gene regulatory network or
there are statistical networks like
co-expression networks
which are very related to things like um
granger causality or
effective connectivity they're
statistically derived
and there might be two elements like two
proteins or two genes or something
that are correlated in their expression
but never touch and those will come up
on the functional connectivity matrix um
or you might have two proteins that
touch and actually are very integra
integrally important for function but
they're not expressed in a correlated
fashion and so
that might lead to them being missed on
the co-expression graph or the

co-activation graph on the brain
and so then somebody might want to say
well what's the what's the real
map of the territory or what's the real
network and that's where this meta
modeling question returns and uh related
to the carving of the joints
yes on the top of figure one we see okay
let's carve out the organism from the
environment
but that's not just a really close line
around the organism
that actually includes some long-range
statistical dependencies
as well as things that might be touching
but not that important for regulating
organismal states
so there's multiple perspectives that
are required
and certainly multiple disciplines that
are required when we want to be
asking how do we design interventions or
systems
given this causal structure the endpoint
isn't just scaffolding out this
infinitely complicated reference map
it's an action-oriented framework that
we want to be talking about multi-skill
models and then also knowing how to
work within them sasha and then anyone
else before we go to figure two
um great point daniel about um
the inter different kinds of
interactions that exist in systems
and whether it's um statistical or like
physical
correlation and
i think the the kind of overarching

theme here is that
you wouldn't know unless you measured it
and
you need to know what sorts of variables
are important
in order to even see statistically if
they're going to be related
so that is where this kind of framework
is really critical because we need to
um measure the relevant
things in our system and if things are
happening at scales that are
too long or too short outside the bounds
of our measurement
then we'll never know nice
we'll do maxwell and then stephen and
then move to figure two
well i just want to pick up a bit on
this meta modeling thing that
that had been mentioned i think it it's
one of the strengths of
to it's one of the strengths of the
framework to on the one hand
um describe like the workings of
organisms using this uh free energy
minimization stuff so
we're sort of uh we're sort of adopting
uh the theory itself the free energy
principle theory itself adopts an
instrumentalist stance on organisms
uh to say well what organisms are
essentially doing is
leveraging the statistical structure of
their bodies in movement which we can
interpret as a generative model as a
joint density over all the system
variables
so using the statistical structure of

their bodies in movement to generate
adaptive patterns of behavior so at that
at that first kind of theory level
there's an instrumentalism uh that's
going on and there's a kind of meta
there's a modeling aspect going on we're
saying that effectively
organisms are sort of in a process of
approximate
bayesian modeling of their the causes of
their sensory
stimuli then there's a kind of meta
level of analysis where we can actually
as scientists write down a generative
model that says well the markov blanket
is here
and then score that model in terms of
the variational free energy
and use a variational free energy
approach to kind of find where
the the boundary is of the variational
free energy minimizing
system so i i kind of take that uh
you know meta aspect to be a strength of
the the framework
nice agreed um we'll go to stephen and
then mel
yeah and so following on from that as
well
the the the piece that sometimes gets
maybe lost here but is in the other
papers when you look at regime of
attention
is that it's not just that the system
cleaves like the whole
the the the mark of blanket gives a
capacity to change how you take your
attention

from the stream that are coming in like
i suddenly feel
my hand then i feel my face and i
suppose one of the things that would be
an example
as well of something that's a very large
scale is
the threat of climate change and i'm
thinking about when i start thinking
about
a glacier in antarctica starting to come
away and go into the
the water and possibly raising sea
levels in 20 years
my physiology changes and my whole
you know when i bring that into my
considerations
the way i take my attention and the way
i tune to a lot of things
around me in the world changes so
you know is it is that has my system
boundary changed or i would
say that it's more about my attention or
integration that's changed so that's
where i think there's a
question around whether the word systems
could be overused sometimes
and it's maybe sustained with this kind
of statistical
dynamic is quite quite useful right it's
uh statistically influencing you
whether you like it or not and then your
awareness and your attentional regime
is something that's almost separate
awesome
mel
oh i just wanted to say um goodbye
because i'm

i'm supposed to be in another girl right
now thank you
for participating now awesome thanks for
having us this is this is lovely
cool so in the last couple of
minutes i just wanted to get all the
figures on
the screen just so we could run through
them let people take the next week to
digest
them a little bit more fully come for
whatever exciting parts they were
interested in
so when i look at figure one i just see
on the top the simplest
framing of system from surrounding and
if you
think that that blurry that's a blurry
boundary then
you're not talking about a system so to
be able to define something is to
uh you know to delimit is to define so
it might not be the most functional
boundary but you can start there and
then you ask well what kinds of
states or interventions connect the
system to the environment just like
muddy was describing
and those are described in this model as
statistical dependencies
and if we had to put english words on
them they would be sensory states and
active states
but ultimately it's really about what
the model actually says
figure 2a goes into a bit more detail
in that action perception internal
external

two-stroke engine as i think about it
this feedback loop from
internal and external states connected
through sense and action
and then the markov blanket is what is
being pierced bi-directionally by sense
and action
and as we've talked about before and in
some of the work on niche construction
one systems internal is another system's
external and so in
some sense what is being uh discussed is
that blanket state because there's some
symmetry between the internal
and the external in that both internal
and external states have their own
self-causing or autonomous dynamics as
well as
influence each other through sense and
action respectively
so figure two is just showing that again
in the context of a human
and then also connecting it a little bit
more closely to the free energy
formalism
of energy minus entropy and so again
another day for talking about the math
there
figure 2b just talks about how those
formalisms of free energy respectively
complexity minus accuracy on the action
bound and then divergence plus surprise
on the perception bound
how the math sort of sketches out
another day
figure three is an illustration of
operational closure so what they note
here is that the black circles

are forming part of an operationally
closed network of self-organizing
processes
and this is something like teleological
closure
like if you're talking about a
sustainable ecosystem you don't want the
sustainable ecosystem to be defined from
your front door to your trash can
you're talking about operational
functional closure
existing at a higher level of analysis
so that means the multi-level
perspective is going to be important
but at one level of analysis or at
multiple
we can ask has operational closure
been achieved have we done enough system
diagramming or
mapping of causal relationships or
dependencies
such that we actually have the system's
dynamics
separated out from dynamics outside of
the system
if i can just jump in the the motivation
for including this
figure uh is part of the argument that
we're making i think that's
controversial is that basically
most of what the inactive approach does
uh we can do with the
active inference approach but in a way
that isn't just
uh like uh well in a way that's
formally backed you know by like a
principled
uh physics-based approach uh and so

we're just discussing here the notion of
operational closure
uh that inactivists use to define an
autopoietic
uh entity like a self-producing entity
and the argument that we make
immediately after is that well
uh active inference
doesn't just doesn't just say there is
this operational closure it tells you
why there is operational closure
nice and also to reference what we're
talking about with
mel just briefly um kant
performed a teleological analysis and
said that the organism
is not just the limits per se of the
epithelium not a physical
boundary but it's actually the
teleologically closed entity
and so that is really consonant with the
con with the idea that the
eusocial insect colony is an organism
not a superorganism by calling it a
super organism
we reify the boundary of organism being
at the epithelia
and people say oh well it's like a
superorganism it's all these little
insect organisms working together
but actually because of the major
evolutionary
shift in eusociality in the ants for
example
the colony is the organism we don't draw
a little single cell organism barrier
around each one of our cells and then
spend all of our time running the math

on the altruism between the nerve cell
and the muscle cell and
the the sex cell and things like that we
just say no
that's the teleological level the liver
is going to make sense
in a broader end and that end is the
organism
and then the question about what the
privileged level of analysis should be
the organismal teleological analysis
says right the liver and the stomach
makes sense at the level of the organism
but in some sense doesn't the organism
only make sense in the context of its
cultural
niche and of its social relationship so
why stop there where should the
teleological closure end where should
the functional closure end
and that's what demands i think a true
multi-level
perspective because there's going to be
questions where the functional boundary
is
the epithelium around the person or
around the single ant nest mate
but there's other times where what we're
talking about is the closure around the
organism as the teleological hole
and that's not always the same thing as
the epithelium yeah richard
yeah i i think this general argument is
is really captured well in the
in the craver and darted mechanistic
philosophical perspective to where
a phenomenon is defined by the entities
and activities that generate it or

produce it and so whatever phenomenon
you just you define whether it's the ant
it's the colony
there's going to be some mechanism that
that generates it and that mechanism is
decomposable
down to whatever you want so the colony
can be decomposed down to the individual
ant
which can be decomposed down to the
various biological structures
and really when you want to move up and
down you're just simply identifying okay
what are the entities and activities
that generate a phenomenon that then
interacts with other phenomenon
entities and activities to create that
right and you build up and down
because i think i think it gets really
dangerous in sociology we do this a lot
to where
there is no individual there's only
society and there's theological
reasoning
where it gets very hard to keep track of
okay well
bob commits a crime not the entire
society that bob's a part of so how can
i drill down on bob
as an entity that i want to talk about
but then recognize that bob is part of a
of a larger structure
and not reducing it to the the group as
the entity in itself
yes it's 100 um related to that
book of helen longino whether you know
was it the arm that committed the
aggression was it the person

or was it the group and so it's funny
i'm sorry
you mentioned that because i mean that's
kind of a symptom in schizophrenia where
a lot of
criminals will say i didn't kill him the
bullet did and
and then logically that makes sense in
terms of that reasoning
but any rational person would say well
no you pulled the trigger so therefore
the bullet
is responsible you know are you
responsible for the bullets pathway
well no no i didn't kill him the bullet
did so that that logic i think when
pushed to its logical extreme
um breaks down pretty quickly but yet
nonetheless i think we do tend to
implicitly rely on it a lot in our
analysis
yep and before we go to stephen it's
again it's those causal stories
guns don't kill people people do well
depending on cognitive diversity that
statement is going to
sound true or not to you and someone
could say well it's not even the bullets
it's actually blood loss or it's not
even blood loss it's reduction in blood
pressure
where does the causal chain end and so
at some point
we need to cut through the descriptive
approach with an action-oriented
approach okay are we gonna ban
bullets guns people where do we stop
there's no single answer just from

identifying the territory
to action and so these are really the
big
questions about causal relationships in
multi-level systems we'll go with
stephen and then muddy
yeah i think the ability to have these
mechanisms
is made possible because of processes
so we think about this in art because
art is seen as a process
that is creating something so in the
same way with the ants
there may be a mechanism by which ants
chew up and masticate pieces of um
wood or mud and there's a mechanism by
which they build the ant colony
and that ant hill has a structural
integrity that allows air to flow
through certain systems of
movement that creates a cooling effect
however
the processes which i think the markov
blankets are working with
are what make the mechanisms possible to
happen
like the ant is following these kind of
statistical dependent
processes to do what it does and in
doing so
it has a mechanism just like if
someone's firing the bullet
there's a process the body's using to
infer
yet the mechanism is something we see
so i think this is these these things
are
kind of entwined but they are slightly

different

agreed muddy and then back to maxwell

um i guess this is where my original

gripe

with philosophy of science came is from

that from an empiricist point of view

for me you know where do you snip the

causal chain

is something that can be empirically

defined you know

if i have an extremely strong coupling

between action a and

action b such that action b doesn't

occur without action a

then move the markup boundary beyond it

right that's that's good so

any time anyone says guns uh you know

bullets don't kill people

um people do or whatever sorry people

don't kill people bullets do i don't

know what the the

spurious phrase is or whatever but you

can say well let me actually empirically

look at this

what percentage of uh and i'm sure there

is like somebody's going to come with an

example oh there was this gun that went

off

itself whatever but statistically

speaking if i threshold that i can find

basically a probability of one

or very very or tending towards this

that the gun was fired by a human being

so i think it's a bit of a when somebody

when somebody presents such a

philosophical argument i'm like well you

know there is an empirical response the

science gives you these empirical tools

to infer these
sorts of things so that's kind of where
my my original gripe about the
phosphorus science coming out
again i've softened on it but i just
wanted to say i think there's an answer
to these things philosophically
and it comes from sampling the data
cool uh maxwell before we finish out the
figures
cool uh well i just want to think about
something that was mentioned in the chat
by richard is that the
so there isn't any i think systematic
work that relates the free energy stuff
to uh neo-mechanistic philosophy
a la craver something i'd like to
discuss with you all next week is
um so as i understand it the
multi-level framework i started off as a
neo-mechanist i have to say like
uh you know the the craver bechtel
orientation was really something that i
was pursuing
and around 2013 2014 i i kind of
spent a lot of time trying to see
whether i was able to
you know use the mechan the
neo-mechanistic framework to articulate
multi-level causation
and my big gripe with uh neo-mechanism
at least in
its most popular renditions is that it
effectively eliminates
uh cross-scale uh causal effects
so there's a part of this that i buy
which is that you know
part of my rejection of metaphysical

levels comes from the neo-mechanist approach uh but i i think that because with the the the story we should discuss this maybe in in another podcast but the their story is something like well

uh what looks like uh you know inter-level causation is really uh is really just uh it's it's just the same causal stream that you're redescribing at another level

one of the things i like about uh the fep formalism is that while it's consonant with the broad outline of uh neo-mechanistic philosophy it does explain the way that you have causal effects that trickle up and down um so i think that's like an outstanding issue that that should potentially be uh discussed in a sort of flag that for the next episode yeah good call um alex k and then i'll close out the figures

all right i was i was actually gonna lower my hand because that maxwell said we could discuss this another time but i guess i

briefly just wanted to say yeah i think if you if you go back to like kant's definition of mechanism which i can't recite but

uh basically he contrasted mechanism with systems that involve top-down sort of causation as well like you know that holes can affect parts

i i would guess you could still get some kind of mechanistic description at each

scale or like approximately or something

but

anyway i think it's an interesting topic

oh yeah yeah just

to be clear my view is that the fep

is broadly consonant with uh and is in

fact an exemplar

of the neo-mechanistic research strategy

as outlined by bechtel and craver

and so on but i think it it does impose

a different metaphysics of levels

uh than the one that's been uh attempted

at least in a few papers

uh that i saw by bechtel and company

like the one called

looking up down and around i think uh

where they essentially say that well

there's no such thing as cross-scale

uh influences they're really just big

things interacting with small things

and uh that's where i i think i strongly

disagree

uh for reasons that hopefully will

unpack uh next week and that are related

to how these multi-scale

systems are actually constructed and i

mean

essentially you've got the same base

system uh

that's acting so it's like i don't know

it's a ladyman and ross style there are

only basically quantum fluctuations or

whatever

but what you're doing is whenever you

define a new scale of interactions

you're adding

a slower and spatially larger set of

constraints on the same

fundamental set of things so there is something like cross-scale talk in the sense that what it is to be a multi-scale system is uh essentially to have components that are constrained by successively higher level uh constraints on the dynamics uh so but i don't think ultimately this this conflicts with the philosophical perspective of neo-mechanism which ultimately at the end of the day just says well there are parts and processes and a mechanism is a coordination and an organization of parts and processes that produces some kind of interesting output that we can study so cool yep these are um alejandra go ahead yeah yeah thank you and just following up this idea uh you're saying max well i i was thinking um if is there any differences in how these hierarchies um in terms of blankets uh work i was wondering um at the cellular level uh maybe belief dating is kind of risky so the cell has to behave like having this generative model but um [Music] it's it's like it's phenotype but it's not learning uh i don't i don't know i'm maybe i i don't have um very clear what i'm thinking but if you go like up to the scale

thinking like in the brain um level
the blanket now um belief of dating it's
it's it's crucial
for for um like this bodily
interactions and effective bodily
interactions
and this coupling with the environment
so um
my question is is there any differences
in terms of
how these micro blankets and blankets of
blankets blankets work
um and this it's also a hierarchy
and how this hierarchy
belief updating works
let's let's let's return to this great
question when we have a lot more time
next week which is about
the similarities and differences with
different systems it's really something
we should
consider because i was gonna say this is
like the question
yeah that we address in this paper in
large part
uh i mean a quick answer to your
question is that formally speaking
there's a kind of self-similar pattern
that that is occurring
across all these different scales right
so you have a non-equilibrium steady
state density
and a generative slash generative model
uh the the and that is uh kind of
directing policy selection in a way that
maintains the
blanket but that's a very very uh like
abstract or information theoretic

description

and in effect uh like basically the
the kind of data that's involved at
every scale the mechanisms involved to
make sure that your data
the data that you're sensing fits your
preferred data distributions all these
things are going to be radically
different from one scale to the next
uh i mean yeah clearly the way that we
update beliefs about ourselves
by having a conversation while it has a
formal
while it may formally be identical on a
kind of process
or information flow level uh to like
what our cells are doing like clearly
this is different right like the the
data streams with which we're dealing
are different the ways that they're
being managed they're different
uh so yeah i mean the the there is a
kind of fractal
self-similar structure uh but it's at
the
it's at the level of the information
flows not so much what makes up
uh so in mechanistic terms the parts are
different but the processes might be
yep and it will be awesome to go into
more detail because there's clearly
differences between
cells that embody their beliefs through
morphology that might be slower time
changing and
brains that can rapidly emulate or at
least you know simulate various
phenotypes being neural states and in a

previous paper we talked about
whether those neural states are
themselves phenotypes
just to quickly run through these last
figures figure four
is an example of auto catalytic closure
so this could be like a metabolic
network or self-replicating nucleotide
sequences or programs that replicate
themselves with or without mutation
and the point here is really to say that
that kind of causal dependency structure
that we saw in figure 3
it might not just be bumper cars or ants
bumping into each other
it could be something like different
types of information
that are connected to each other and so
as we start to think about different
spatial temporal scales and all these
different mechanisms that exist
we want to find those invariances at a
structural
and dynamical levels that help us study
this diversity of systems
using a common formal framework figure
five
is something we already looked a little
bit at but it is this fractal idea
zooming in on the fractal broccoli and
basically
the left side has uh sensory and action
states surrounding three gray nodes
but that's itself zoomed in from a swarm
of
similar agents and so in a tissue maybe
it's very patterned
like in the skin cells the boundaries

between the cells are very stable
so they can have adhesive molecules that
make it a functional
skin sheet and they're still exchanging
information
in that grid topology but also maybe the
swarm on the right side consists of
moving
subunits so like steven was talking
about with a swarm
and at that point especially it is
important to have a statistically driven
understanding of causal relationships
rather than one that's just around
drawing a dashed line around the system
as we visually see it
and then just lastly figure six which i
think will be perfect
to save for later especially because it
in some ways does
connect this action inference
generalization
to specifically morphogenesis which is
an example of a phenotype decision
making that people might not
always expect to be put in the same
category along with
what it is our brains or cpus are doing
but i think that's a perfect jumping off
point for
next week when we uh yeah nice
and going through this is going to
hopefully address
alejandro's question uh i think so yeah
this is a good
jumping off point i think perfect and we
have a few questions
just prepared for next week but we're

not gonna go to them now
we will talk about that next time that
concludes
the main section and we'll also just
have anyone can raise their hand and
give a final point but i just wanted to
say that
we'll provide a follow-up form to our
live participants i'll put it in this
jitsi chat
as soon as someone starts to give a
final thought
any feedback suggestions or questions
from
our listeners participants are always
welcome
of course everyone's engagement is
welcomed and people can
stay in communication with us so if
anyone has their
hand raised i'll be happy to hear their
first thoughts while i post the link
nobody's hand is raised but thanks again
for the continued engagement with our
work
um it means a lot and i think this is
extremely useful
uh like both i think for for us as a
little community working through these
things and
uh you know for the broader uh audience
of
the free energy active inference stuff
so i'm very grateful thank you
very much awesome um muddy
um raised your hand and if anyone else
wants to give any
final thoughts what they reflected on

from today's discussion what they'd be
excited to hear about in next week
discussion

uh i was just going to say um thanks sir
this is really really great i think um
first of all just on a personal level
it's great to get back involved with
people who are much cleverer than i am
and be surrounded by this this new
knowledge but secondly i can really
start to sense the snowball effect in
the active inference community
it's only a matter of time before it
really leads over into industry
it's a really really exciting time to
get involved in the space
thank you all that's really good cool
richard and anyone else

freely raise your hand yeah i i think
this is really interesting and i think
there's there's so many different points
of connection and interaction with so
many different disciplines
and um i think one of the difficulties
is the barrier to entry is just so high
in terms of the terminology the
mathematics

to try and rapid your head around these
things but once you begin to understand
it conceptually i think people can start
go okay

i know what this is and i can talk about
this and so i think one thing to
maybe you know discuss moving forward is
are there ways to kind of get
almost sandbox working groups where
people can kind of start
really empirically playing around with

ideas and data sets
um that kind of emerge naturally out of
these conversations because
you know cold calling somebody and
saying hey i've got this idea can you
explain to me what a marcotte blanket is
let's work together people are like
no son i got things to do i'm sorry
i think you know there's a way that we
can kind of merge from these really
great productive working
our discussions into maybe something
where people start producing
actual collaborations across disciplines
i think would be a really fantastic
thing
awesome i i just want to reiterate also
uh
that there's this reading group uh
that's ongoing
on the carl firstens monograph so for
those of you who are
interested in actually uh you know uh
surmounting the barrier to entry as it
were uh we're
we we've recorded uh the the chats and
everything so if you do want to join
just uh drop me a dm
um and yeah we've grown to
this is originally going to be like me
and eight buddies just reading through
the monograph and we've grown to a
community of almost 100 scholars
uh yeah following this uh
like remotely with the recordings
available and we're going to turn
these recordings heavily edited and
everything

into pedagogical material a kind of like
video based
a companion to a free energy principle
for a particular physics
uh so yeah if anyone is interested uh
the group is kind of semi
loosely open um to especially the
to people who are doing active research
uh in the tradition
so uh yeah drop me a dm and i'll all
hook you up cool stephen and then anyone
else if their last thought
comes to mind
yeah well just to reiterate yeah it's
really useful i like to see everyone
nibbling at this from different
directions are sort of it it shows a lot
the questions everyone's asking
it's like because otherwise you're
thinking it's just me that's thinking
this
and sometimes it is and then sometimes
it's not so it's kind of good to know
which is which
because sometimes it's not clear it's
just a figure of my imagination as the
only other thing i thought might be cool
to bring
is that new paper that came out on
reciprocal active influence that's just
come out
might be interesting at some point to
think about because that might open up
some another level of
development from this paper a very
interesting paper by matt
sims uh at edinburgh who
um i'm i've been very excited about this

stuff

so matt sims developed a um
an account of symbiosis that draws on
the multi-scale integration framework
that we've

uh developed uh where he
talks about bi-directional uh
multi-scale integration so not just
integration

of the parts of of one biological
organism but the integration
of effectively a symbiotic union of
organisms with different phenotypes uh
the paper uses the uh i think it's the
bobtail squid

that uh gets colonized by bacteria and
so

yeah it's a very very interesting paper
and i highly recommend it

cool cool very interesting sasha

yeah i just wanted to reiterate what
stephen just mentioned

um it's great to have this ongoing
conversation

and to hear how different people phrase
the questions that i think we're all
thinking so i'm really looking forward
to next week's discussion

um and kind of uh seated with the
question that alejandra
stated earlier perfect well

thank you all for participating the
google form for you to provide any
feedback for the live participants is in
the chat

for people who are watching this live or
in replay please provide us your
feedback so we can make this

better on-ramp for active inference the
community the ideas
and ultimately the projects too so it's
just
great stuff another one in the books 5.1
for active and i'm going to terminate
the live stream thanks so much for
listening
thanks